

EV-Based Evacuation Readiness — Indicator Scoring Tables (Final Workshop)

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1 Reminder: How scoring ranges affect the values?

The scoring ranges determine how a value is classified as Bad, Medium, Good, or Excellent. Adjusting these ranges changes both the span and the overlap of each category. This means that in some cases, a single value may fall into two categories at once (for example, partly “Good” and partly “Bad”), depending on how the ranges intersect.

Figure 1 shows the initial range. Figure 2 illustrates a new range where the Bad values go from 0 to 1.2. With this change, there is no longer any overlap between the two zones.

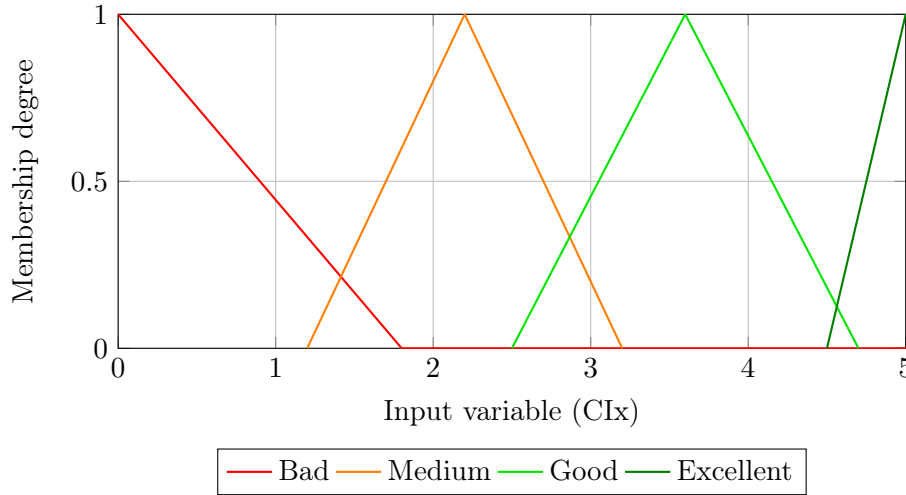


Figure 1: Triangular membership functions for initial input variable CIx

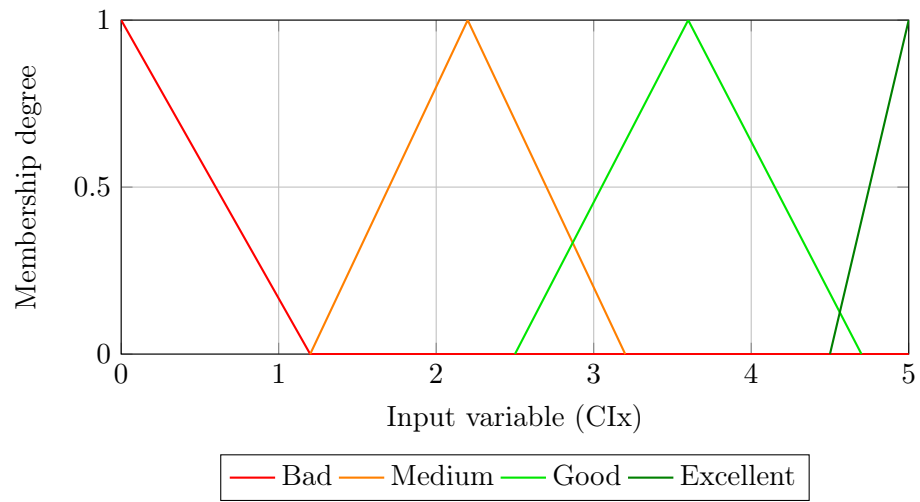


Figure 2: Triangular membership functions for new input variable CIx

2 Community Engagement and Preparedness

Table 1: Scoring table for CEP1: EV User Communication and Readiness during Emergencies

Level	Criteria met	Description
Bad (0–1.8)	No communication strategy	No documented communication to EV users before or during emergencies. No guidance on charging, SoC levels, or safety procedures.
Medium (1.2–3.2)	Basic outreach	Some communication exists (e.g., website, flyer, or one-time campaign) but limited in scope or reach. No systematic updates.
Good (2.5–4.7)	Multi-channel outreach	Regular information provided to EV users (social media, radio, billboards) with basic preparedness guidance (e.g., maintain SoC >30%). Limited feedback mechanisms.
Excellent (4.5–5)	Comprehensive communication plan	Continuous, institutionalized outreach program covering SoC reserve policies, charging safety, and emergency procedures. Feedback is collected and used to improve communication.

Table 2: Scoring table for CEP2: Emergency EV Charging Access Agreements and Public-Private Coordination

Level	Criteria met	Description
Bad (0–1.8)	No agreements	No evidence of agreements or coordination with private charging operators.
Medium (1.2–3.2)	Informal discussions	Preliminary discussions with utilities or private operators, but no formal agreements or protocols in place.
Good (2.5–4.7)	At least one MOU signed	Formalized agreement with at least one charging operator, including limited emergency access. Activation procedures partially defined.
Excellent (4.5–5)	Multiple agreements + integration	Multiple MOUs signed, protocols for activation established, real-time data sharing integrated with emergency management systems.

Table 3: Scoring table for CEP3: Community Planning for ZEV-Based Emergency Evacuations

Level	Criteria met	Description
Bad (0–1.8)	No EV evacuation plan	No dedicated plan for ZEV-based evacuations.
Medium (1.2–3.2)	Draft or limited plan	Early draft of a plan, addressing EVs superficially, with limited coordination or resources.
Good (2.5–4.7)	Published plan with EV considerations	Community has an official plan that identifies EV owners, charging points, and routes. Includes some provisions for staggered evacuations or mobile charging.
Excellent (4.5–5)	Comprehensive and tested plan	Plan fully operational, includes staggered strategies, mobile charging deployment, and coordination with agencies. Regularly updated and tested through drills.

Table 4: Scoring table for CEP4: Plans to Conduct Emergency Response for EVs Left Behind

Level	Criteria met	Description
Bad (0–1.8)	No SOPs or plans	No plan or procedures for EVs left behind during evacuations.
Medium (1.2–3.2)	Preliminary awareness	Awareness of EV fire/contamination risks but only informal or preliminary guidance.
Good (2.5–4.7)	Basic SOPs and training	Standard Operating Procedures exist for isolation and firefighting, with some first responder training and limited monitoring protocols.
Excellent (4.5–5)	Comprehensive emergency response	Full SOPs in place, responders trained, monitoring protocols for air/water contamination implemented, and post-incident handling procedures established.

3 Charging Infrastructure

Table 5: Scoring table for CI2: Community awareness of charging energy needs for evacuation

Level	Criteria met	Description
Bad (0–1.8)	No evidence of estimation	No documented estimation of power/energy needs for EV-based evacuation. No consideration of load surges or energy constraints during evacuation.
Medium (1.2–3.2)	1 factor considered (rough estimate)	Community has begun estimating total evacuation energy needs, but the approach is limited or not scenario-based. No formal tool use or methodology. Estimates are qualitative or rely on rough calculations.
Good (2.5–4.7)	2 to 3 key factors estimated + partial use of tools or structured method	Community has quantified energy needs based on some variables (e.g., number of EVs and charging needs). Basic modeling or tool use (even if informal or partial). Results inform planning discussions but are not yet formalized.
Excellent (4.5–5)	Estimations conducted across at least 4 key factors + modeling tool used + integrated into planning	The community has estimated evacuation energy needs by scenario, including at least 4 of the following: number of EVs, evacuation distance, initial battery state, number of stations, time to evacuate. Uses tools such as EVI-X Toolbox or equivalent. Estimates are documented and considered in emergency or energy planning documents.

Table 6: Scoring table for CII

Level	Criteria met	Description
Bad (0–1.8)	No relevant action identified	No documentation, agreements, or cost analysis available. No indication of planning or coordination related to mobile charging in emergency contexts.
Medium (1.2–3.2)	1 recommended action fulfilled (at least partially)	Preliminary documentation or planning exists (e.g., list of available assets or potential sites). No formal agreements or economic justification completed. Communication mechanisms are still under development.
Good (2.5–4.7)	2 out of 3 recommended actions fulfilled	Deployment strategy exists and at least one of the following is fulfilled: formal agreements or cost-benefit analysis. Initial work on communication or coordination is underway but not fully formalized.
Excellent (4.5–5)	All 3 recommended actions fulfilled + integrated into emergency or resilience planning.	Documented deployment strategy (e.g., unit quantities, deployment locations, responsible entities). Signed MoUs or agreements with site hosts and/or emergency response agencies. Cost-benefit analysis comparing mobile and fixed EV charging solutions. Real-time communication system in place to inform residents about charger availability during emergencies. Clear integration into emergency response or transportation planning.

Table 7: Scoring table for CI3: Readiness of Public EV Charging Infrastructure

Level	Criteria met	Description
Bad (0–1.8)	Number of EVs/Number of EV chargers > 40 < 50% of critical zones covered Severe/system-wide congestion risk No sensitivity analysis or only basic assumptions No mitigation plan	Community lacks modeling or implementation strategies. Critical gaps in infrastructure, coverage, and preparedness planning. Major risk during evacuations.
Medium (1.2–3.2)	Number of EVs/Number of EV chargers between 21–40 50% coverage of critical zones Moderate congestion in several areas Some basic scenario testing Some ideas for mitigation, not yet formalized	Community has initiated planning and simulations but lacks full integration or detailed response actions. Readiness remains partial.
Good (2.5–4.7)	Number of EVs/Number of EV chargers between 11–20 Coverage > 75% of critical zones Limited congestion at max 1–2 sites Partial sensitivity analysis General recommendations outlined	Community demonstrates solid planning with moderate modeling and preparedness. Some scenarios explored, infrastructure generally well-distributed.

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Table 7 (continued)

Level	Criteria met	Description
Excellent (4.5–5)	Number of EVs/Number of EV chargers 10 in all zones 100% coverage of critical zones No congestion under evacuation simulations Full sensitivity analysis with contingency plans Detailed and budgeted mitigation plan	Community has robust planning across all dimensions, based on modeling, sensitivity analysis, and mitigation strategies. Fully integrated into emergency response.

Table 8: Scoring table for CI4: Community has considered the charging of electric heavy-duty vehicles which are either helping in charging (mobile charging units) or directly in evacuation

Level	Criteria met	Description
Bad (0–1.8)	No evidence of consideration or estimation	No plans or data on ESV charging needs, vehicle availability, or infrastructure designation in emergency evacuation scenarios.
Medium (1.2–3.2)	Only one factor is considered (e.g., availability of heavy-duty EVs or charger mapping)	Some preliminary steps are taken (e.g., knowing how many ESVs are in the region), but without integration into energy planning or real-time coordination.
Good (2.5–4.7)	At least two of the three dimensions are addressed (e.g., charger reservation + estimation, or mapping + surge estimate)	ESVs are integrated into evacuation plans, with surge energy needs roughly estimated or some infrastructure reserved, but not fully modeled or coordinated.
Excellent (4.5–5)	All three key dimensions are considered: surge load estimation, real-time coordination tools, and ESV-specific charger planning.	Community models peak ESV energy demand, has mapped ESV routes and locations, designated exclusive charging capacity, and maintains real-time updates for public routing and emergency responders.

Table 9: Scoring table for CI5: Charging Management and Emergency Priority Access for EVs

Level	Criteria met	Description
Bad (0–1.8)	No planning, prioritization mechanisms, or coordination efforts exist for charging access in emergencies.	Charging access is unmanaged during evacuations. No protocols exist to prioritize critical ZEV fleets. Charging stations operate on a first-come, first-served basis, risking congestion, delayed emergency responses, and grid instability.
Medium (1.2–3.2)	Awareness of the need for charging access control during emergencies exists, but no formal implementation. May include discussion of priorities or rough estimates of traffic/EV loads.	Planning is at an early stage. The community may acknowledge the need for priority ZEV access or demand management, but lacks real-time coordination tools, structured protocols, or signed agreements. Implementation remains conceptual.
Good (2.5–4.7)	Partial implementation of key elements: Informal agreements or operational practices. Initial capacity planning or prioritization protocols. Tools or communication methods exist but are not fully coordinated	Some coordination tools or protocols (e.g., time limits, dedicated ports) are in place. Priority access may exist in practice, especially for emergency fleets, but integration into formal emergency frameworks is limited or incomplete.

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Table 9 (continued)

Level	Criteria met	Description
Excellent (4.5–5)	A comprehensive and documented strategy exists including: - Capacity planning (load estimation and demand forecasting) - Priority protocols (emergency vehicles and vulnerable groups) - Real-time tools (apps, signage, digital queue) - Backup systems in place - Formal MOUs with public/private charging providers	The community uses predictive tools (e.g., StreetLight Data), defines and enforces priority access for emergency ZEVs (e.g., ambulances, fire trucks), and implements scheduling and real-time access coordination. Measures are formally embedded in emergency and resilience plans.

4 Mobile and Backup Power

Table 10: Scoring table for MB1: Planning and Deployment of Mobile Charging and Backup Power Units During Emergencies

Level	Criteria met	Description
Bad (0–1.8)	No evidence of planning	No strategy for mobile charging or backup power units. No agreements or deployment considerations.
Medium (1.2–3.2)	Preliminary awareness	Some awareness of the need for mobile charging or backup units, but no concrete plans. No defined protocols or activation triggers.
Good (2.5–4.7)	Inventory and partial strategy	Community has identified mobile units and possible sites. Preliminary deployment strategy exists, but not fully operationalized.
Excellent (4.5–5)	Comprehensive deployment strategy	Full deployment strategy in place with MOUs, risk-based site selection, activation protocols, and coordination with utilities. System tested in exercises or drills.

Table 11: Scoring table for MB2: Large EV Vehicles with V2X Mobilized as Mobile Chargers

Level	Criteria met	Description
Bad (0–1.8)	No evidence of consideration	No planning for V2X vehicles to support emergency power.
Medium (1.2–3.2)	Preliminary awareness	Some knowledge of V2X-capable vehicles in the region, but not included in emergency planning.
Good (2.5–4.7)	Inventory and draft agreements	Available V2X vehicles mapped and preliminary coordination with fleet owners considered. Integration into emergency planning is partial.
Excellent (4.5–5)	Full integration into emergency response	Comprehensive V2X plan with MOUs, dispatch protocols, and prioritization of critical facilities. Vehicles pre-equipped and ready for emergency deployment.

Table 12: Scoring table for MB3: Critical EV Charger Locations Identified and Availability of Backup Power Confirmed

Level	Criteria met	Description
Bad (0–1.8)	No identification	No mapping of critical EV charging sites or backup power availability.
Medium (1.2–3.2)	Preliminary awareness	Some vulnerable chargers identified, but no backup systems assessed.
Good (2.5–4.7)	Priority mapping with partial backup	Priority charging sites mapped and some backup power solutions identified. Real-time monitoring not included.
Excellent (4.5–5)	Full backup power integration	Critical sites mapped, equipped with backup systems (batteries, microgrids, generators). Real-time monitoring and redundancy in place.

Table 13: Scoring table for MB4: Resilience of Power Systems at Remote Locations

Level	Criteria met	Description
Bad (0–1.8)	No strategy	No consideration of power system resilience in remote areas.
Medium (1.2–3.2)	Basic awareness	Acknowledgment of risks in remote areas, but no backup systems in place.
Good (2.5–4.7)	Partial deployment	Some standalone backup systems or ESVs available in remote areas, but coverage is incomplete.
Excellent (4.5–5)	Comprehensive resilience strategy	Comprehensive plan including backup systems, ESVs, and mobile chargers deployed to support rural evacuations. Integrated into emergency management.

Table 14: Scoring table for MB5: Alternative Fuel Recharging Options (Hydrogen, Biodiesel, etc.)

Level	Criteria met	Description
Bad (0–1.8)	No options identified	No planning for hydrogen, biodiesel, or other alternative fuels in evacuation scenarios.
Medium (1.2–3.2)	Preliminary awareness	Alternative fuel options mentioned, but not mapped or assessed.
Good (2.5–4.7)	Mapping and feasibility	Mapping of alternative fuel recharging assets, with initial feasibility studies or agreements.
Excellent (4.5–5)	Integrated alternative fuel strategy	Hydrogen and alternative fuel systems fully integrated into evacuation plans, supported by MOUs with suppliers and included in emergency protocols.

Table 15: Scoring table for MB6: Policies and Incentives to Support Backup Power Investment

Level	Criteria met	Description
Bad (0–1.8)	No supportive framework	No policies or incentives for backup power systems at EV charging infrastructure.
Medium (1.2–3.2)	Initial discussions	Some local or state-level discussions, but no concrete programs.
Good (2.5–4.7)	Programs exist	At least one local or state program exists, providing incentives or requiring backup systems for EV infrastructure.
Excellent (4.5–5)	Comprehensive framework	Multiple incentive programs, mandates, and equity-focused policies in place. Fully integrated into energy resilience and emergency planning.

5 Transportation

Table 16: Scoring table for T1: EV-ready and resilient evacuation routes

Level	Criteria met	Description
Bad (0–1.8)	No EV-specific route planning	No identification or mapping of EV-suitable evacuation routes. No consideration of charging availability, road constraints, or hazards.
Medium (1.2–3.2)	Draft mapping (limited EV focus)	Preliminary routes identified with limited awareness of charging assets or hazards. No risk assessment or contingency measures.
Good (2.5–4.7)	Mapped routes + partial resilience measures	EV-friendly routes mapped and aligned with known charging sites; basic risk screening performed. Some contingency measures (e.g., temporary chargers, shoulder use) noted but not fully operationalized.
Excellent (4.5–5)	Operational, resilient routes	Routes are validated, include contraflow or staged departures where relevant, and integrate backup power at critical chargers. Real-time monitoring, incident response, and public communication protocols are in place.

Table 17: Scoring table for T2: Integration of EV charging awareness into navigation and traveler information

Level	Criteria met	Description
Bad (0–1.8)	No integration	No use of apps or traveler information systems with EV charging data for evacuations.
Medium (1.2–3.2)	Basic awareness (static info)	Static lists/maps of chargers referenced; no real-time status, reservations, or outage alerts.
Good (2.5–4.7)	Real-time data for key apps	At least one major navigation app (or agency system) integrates real-time charger availability and routing with charging stops. Limited redundancy if connectivity fails.
Excellent (4.5–5)	Comprehensive, redundant traveler info	Multi-platform integration (apps, 511, agency dashboards) with real-time charger status, reservations where applicable, and fallback communications (e.g., radio/SMS signage). Aligned with emergency operations.

Table 18: Scoring table for T3: First-responder and roadside assistance protocols for EVs during evacuations

Level	Criteria met	Description
Bad (0–1.8)	No EV-specific protocols	No guidance, training, or equipment for EV incidents (fires, depleted battery, towing).
Medium (1.2–3.2)	Initial guidance or training	Basic awareness materials or ad-hoc training provided; limited access to suitable equipment.
Good (2.5–4.7)	Formal SOPs + partial equipment	SOPs in place for EV fires, isolation, and safe towing; some mobile charging/boost capability or EV-capable tow trucks available. Coordination with transportation and emergency agencies initiated.
Excellent (4.5–5)	Comprehensive EV response system	Full suite of SOPs, trained personnel, equipped tow/assist fleets, and mobile charging units. Integrated dispatch and communication with traffic operations and emergency management.

6 Equity

Table 19: Scoring table for E1: Equity-driven readiness for underserved communities

Level	Criteria met	Description
Bad (0–1.8)	No equity provisions	No identification of disadvantaged areas; no targeted measures for access to charging, mobility, or backup power.
Medium (1.2–3.2)	Awareness + preliminary actions	Underserved areas identified; pilot actions considered (e.g., temporary chargers or outreach) but not systematic.
Good (2.5–4.7)	Targeted programs underway	Dedicated measures implemented (mobile/backup power, public EV access, outreach) in priority communities; monitoring begins.
Excellent (4.5–5)	Integrated equity strategy	Sustained investments and policies ensure resilient charging, mobility options, and communications in underserved areas. Ongoing monitoring with clear equity metrics and course correction.